

EC2 Instance

Creating a Linux Virtual Machine

Presented by: Juveriya Nadaf

Subject: Cloud Computing



Introduction to Cloud Computing

What is Cloud Computing?

Cloud computing is the delivery of computing resources such as servers, storage, networking, databases, software, and analytics over the internet. It allows users to access resources without owning physical infrastructure.





Introduction to Amazon Web Services (AWS)

AWS is the world's leading cloud platform launched by Amazon in 2006. It provides more than 200 fully featured services from data centers worldwide.

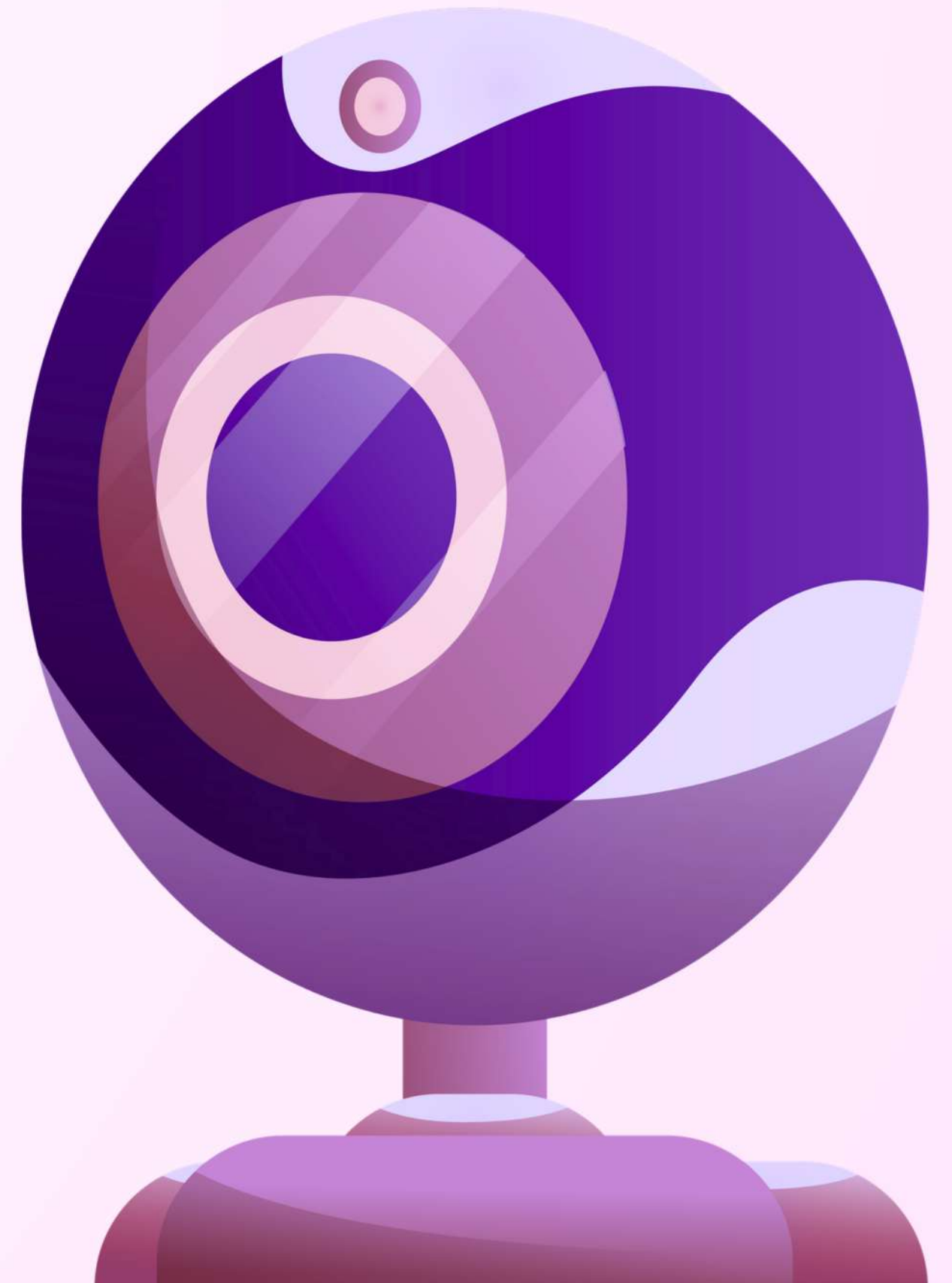
- **Popular AWS Services**

1. EC2 – Virtual Servers
2. S3 – Object Storage
3. RDS – Relational Database
4. Lambda – Serverless Computing

What is Virtual Machine

A Virtual Machine (VM) is a virtual computer created using software. It behaves like a real computer and has its own operating system, CPU, memory, storage, and network, but it uses the hardware resources of a physical computer.

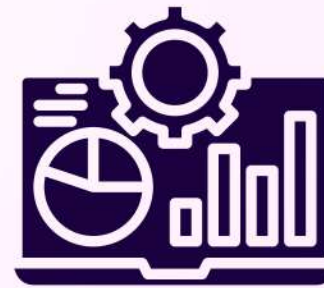
A VM can run applications, store files, and connect to the internet just like a physical computer. Multiple virtual machines can run on a single physical computer, and each VM works independently without affecting the others.



Advantages



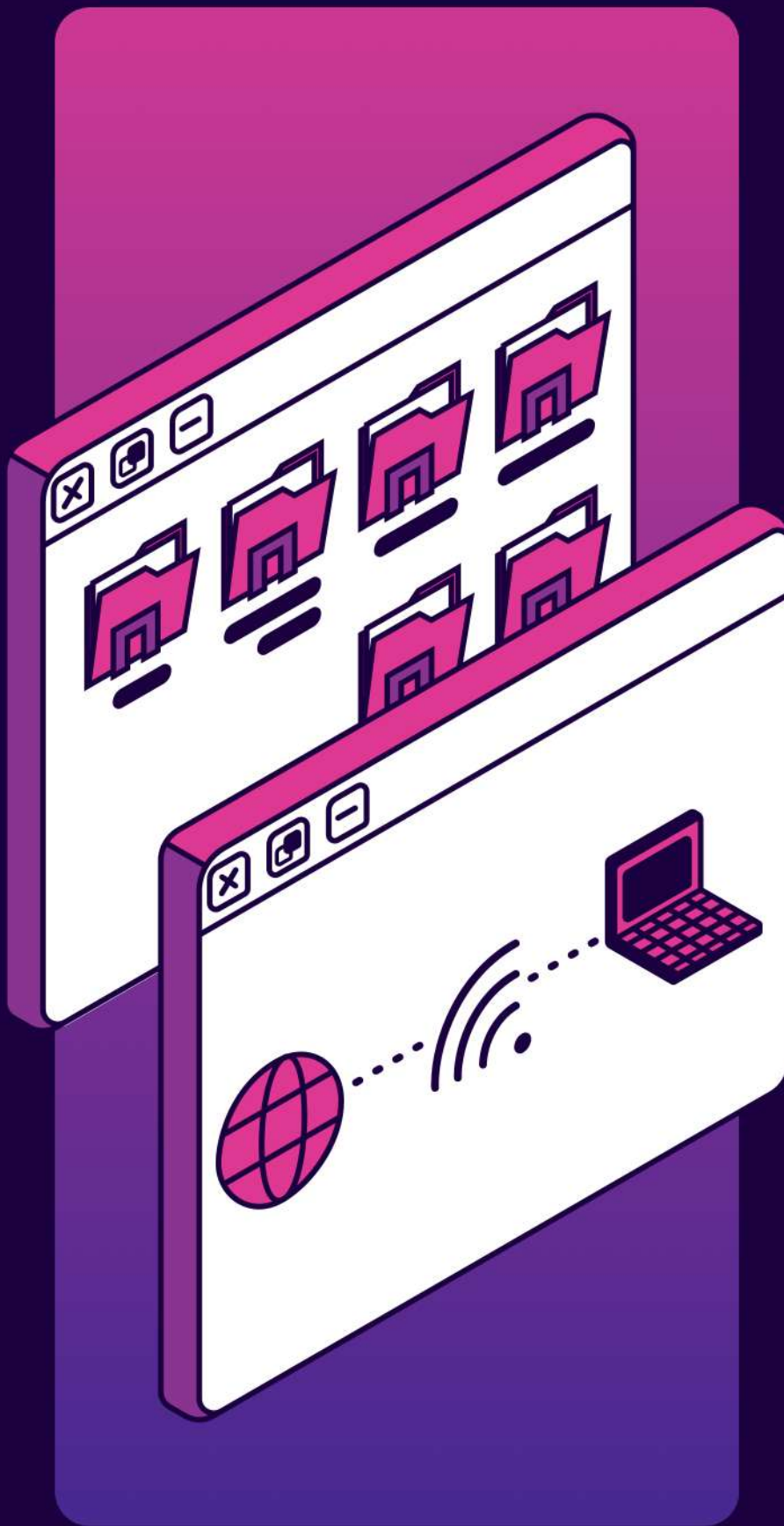
Reliable



Highly Secure



Flexible pricing



What is Amazon EC2

Amazon EC2

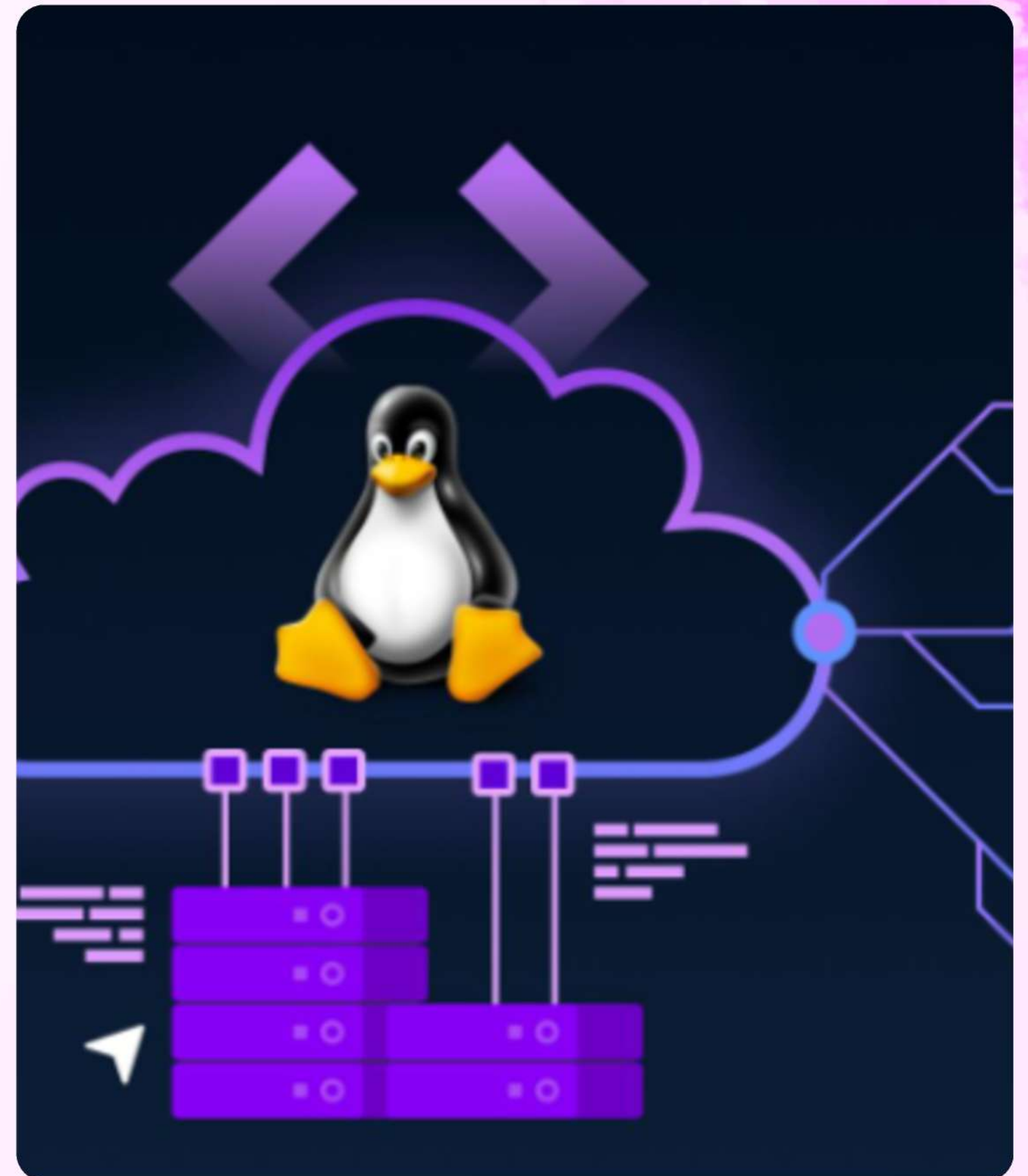
Amazon EC2 (Elastic Compute Cloud) is a cloud computing service provided by Amazon Web Services (AWS) that allows users to create and manage virtual servers (called instances) on demand. It enables you to run applications in the cloud without purchasing or maintaining physical hardware.

Features of Amazon EC2

- Virtual Servers (Instances) – Launch and manage virtual machines in the AWS cloud.
- AWS Documentation
- Scalability – Increase or decrease computing resources based on demand using Auto Scaling.
- Amazon Web Services, Inc. +1

Linux Virtual Machine

A Linux Virtual Machine (Linux VM) is a software-based computer that runs the Linux operating system on virtualized hardware. It behaves like a physical computer but shares the resources (CPU, memory, storage, and network) of a host machine through a hypervisor.





Supported Distributions

Amazon Linux

Ubuntu

Red Hat Enterprise Linux

Debian

SUSE Linux

Why Linux

Open-Source Operating System

Linux is an open-source operating system, which means its source code is freely available. Users and organizations can modify and customize it according to their requirements.

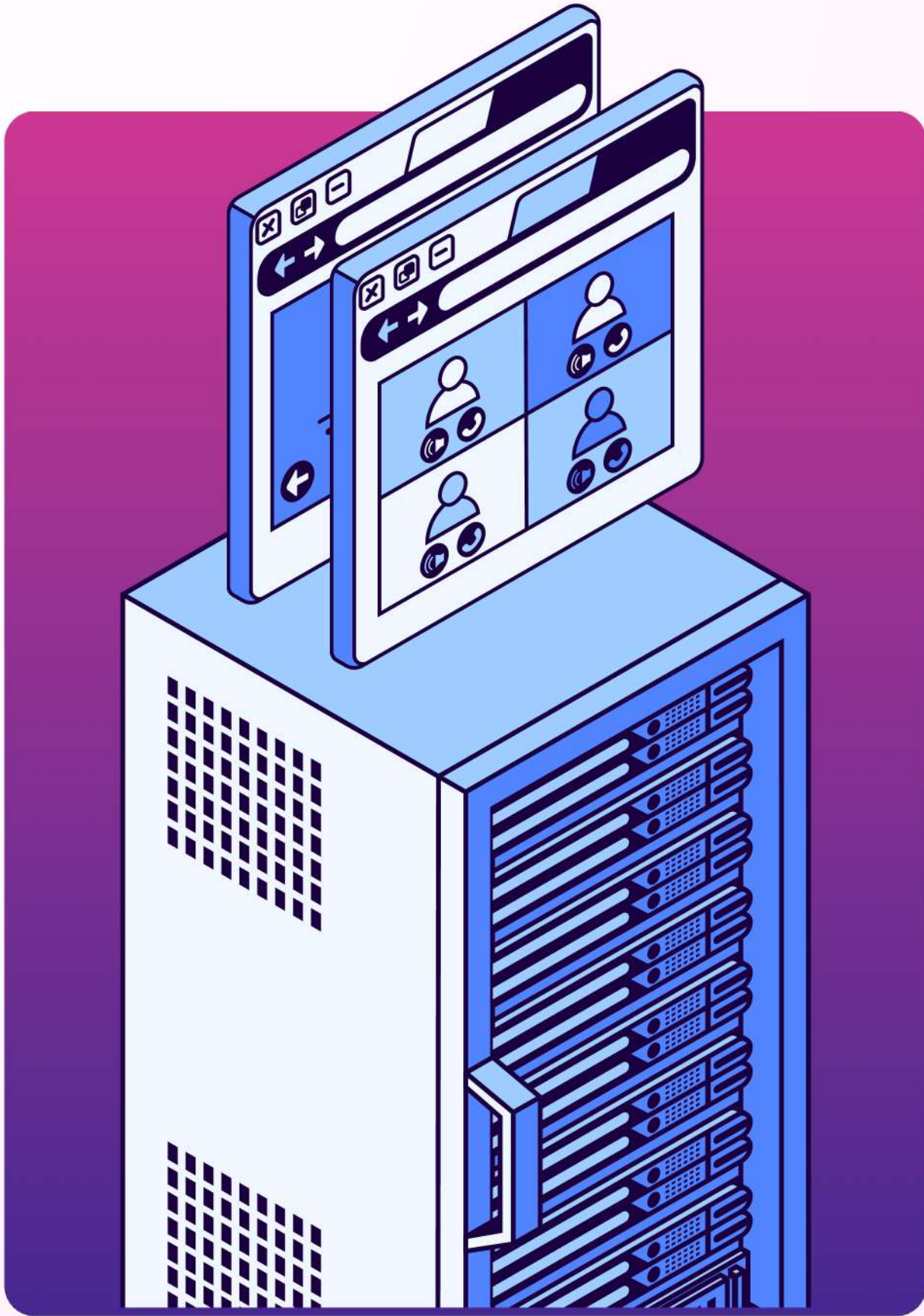
There are no licensing costs for most Linux distributions, making it economical for businesses.



High Performance

Linux uses fewer CPU and memory resources than many other operating systems. It provides fast boot times and efficient resource management. It is ideal for running web servers, databases, and cloud applications.

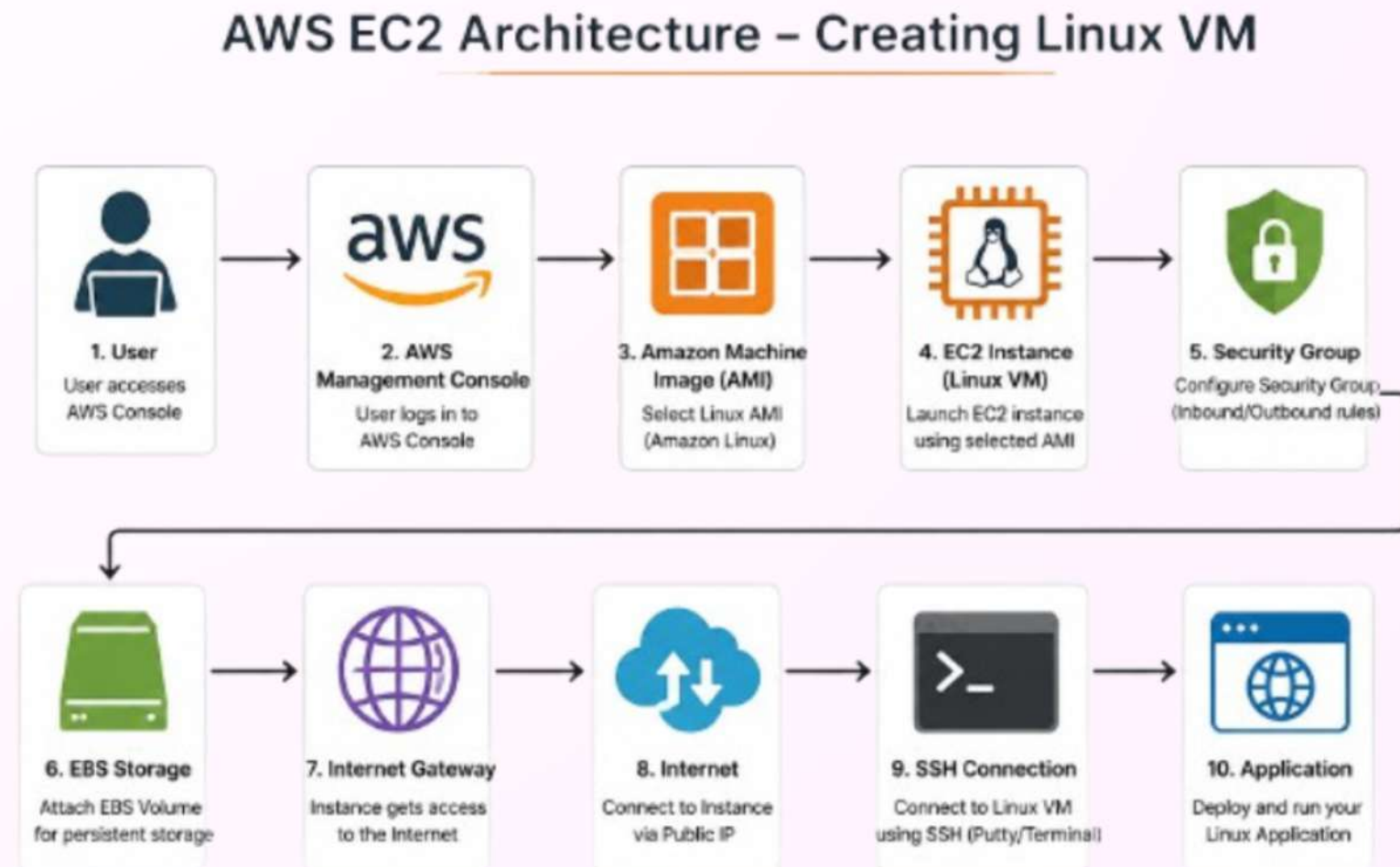




EC2 Architecture

Amazon EC2 (Elastic Compute Cloud) Architecture is the structure that shows how EC2 instances interact with AWS networking, storage, and security services to run applications in the cloud.

Diagram Flow:



Components:

1. AMI (Operating System Image)
2. Instance Type
3. Key Pair
4. Security Group
5. Elastic Block Store (EBS)
6. Elastic IP

Steps to Create a Linux VM

- Sign in to AWS Console
- Open EC2 Dashboard
- Click Launch Instance
- Enter instance name
- Choose Amazon Linux AMI
- Select t2.micro (Free Tier)
- Create Key Pair
- Configure Security Group
- Review settings
- Launch Instance



Security Group

A Security Group acts as a virtual firewall that controls inbound and outbound traffic.

Port Protocol Purpose

Type	Protocol	Port	Source
SSH	TCP	22	Your IP Address
HTTP	TCP	80	Anywhere (0.0.0.0/0)
HTTPS	TCP	443	Anywhere (0.0.0.0/0)

Basic Linux Commands

Command	Description
pwd	Current directory
ls	List files
cd	Change directory
mkdir	Create directory
touch	Create file
cat	Display file contents
cp	Copy files
mv	Move/Rename files
rm	Delete files
sudo yum update	Update Amazon Linux
sudo apt update	Update Ubuntu

```
Y SSH-2 private key)
Using username "ec2-user".
Authenticating with public key "Juveriyaa"

#
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~~\#/ https://aws.amazon.com/linux/amaz
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[ec2-user@ip-172-31-9-86 ~]$ mkdir Student
[ec2-user@ip-172-31-9-86 ~]$ cd Student
[ec2-user@ip-172-31-9-86 Student]$ touch Student.txt
[ec2-user@ip-172-31-9-86 Student]$ ls Student.txt
Student.txt
[ec2-user@ip-172-31-9-86 Student]$ vi Student
```

Advantages of EC2

- Cost-effective pricing
- Easy scalability
- High availability
- Secure environment
- Fast deployment
- Flexible configurations
- Global infrastructure
- Integration with AWS ecosystem

Real-World Applications

- Website hosting
- E-commerce platforms
- Cloud laboratories
- DevOps pipelines
- Artificial Intelligence
- Big Data Analytics
- Mobile app backend
- Online gaming servers
- Disaster recovery

Conclusion

This presentation introduced the basics of cloud computing and the process of creating and managing a Linux Virtual Machine (VM) using Amazon EC2. It covered launching and configuring an EC2 instance, connecting securely through SSH, performing basic Linux operations, and understanding security groups, key pairs, and storage. We also explored the key benefits, real-world applications, and best practices of AWS EC2. Overall, this presentation provides a strong foundation for using EC2 to deploy and manage secure, scalable cloud-based virtual servers.

THANK-YOU